Complex Numbers

Basic arithmetic with complex numbers

1 Given $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$, find $z_1 + z_2$ and $z_1 - z_2$.

2 Find $z_1 z_2$ for $z_1 = 2 + 3i$ and $z_2 = 4 - i$.

The complex conjugate and division

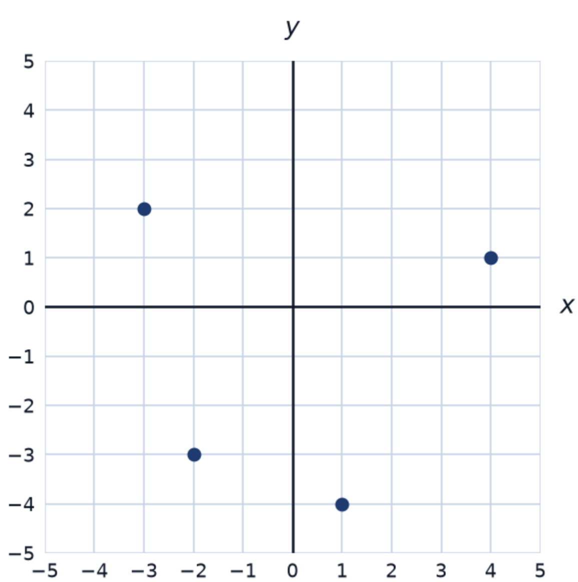
3 Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.

4 Simplify $\dfrac{3+i}{2-3i}$ by multiplying by the conjugate of the denominator.

Modulus and the Argand diagram

5 Find the modulus $|z|$ of $z = 5 - 12i$.

6 The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).



a Read off the complex number plotted at $(4, 1)$ in Cartesian form.

b Find the modulus of the complex number plotted at $(-2, -3)$.

Solving equations with complex roots

7 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

8 One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.

9 Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.

Converting to polar (modulus-argument) form

10 Write $z = \sqrt{3} + i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.

11 Convert $z = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ to Cartesian form.

Multiplying and dividing in polar form

12 Given $z_1 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

13 Given $z_1 = 5\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)$ and $z_2 = \cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

14 Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.

De Moivre's theorem

15 Use De Moivre's theorem to find $\left[3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.

16 Use De Moivre's theorem to find $(1 - i)^5$, giving your answer in Cartesian form.

Finding roots using De Moivre's theorem

17 Find the three cube roots of $z = 27\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.

18 Find all four fourth roots of -16 (i.e. solve $z^4 = -16$), giving arguments in $[0, 2\pi)$.

A well-designed word problem: AC circuit impedance

19 This question has three parts. An AC circuit has impedance $Z = R + iX_L$, where $R = 8\,\Omega$ (resistance) and $X_L = 6\,\Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I = \frac{V}{Z}$, where $V = 100$ volts (real). Find I in the form $a + bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.